
Thesis Proposal
Doctoral Program
<Technology Management>

AI Adoption in
Cambodian Family Businesses

Haiyan Li
<lilian.li@camtech.edu.kh>

Advisor(s):
<Prof. Dr. Chhem Kieth Rethy>

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School for Graduate Studies
CAMBODIA UNIVERSITY OF
TECHNOLOGY AND SCIENCE

Abstract

In emerging economies like Cambodia, family businesses form the backbone of economic growth, contributing significantly to GDP and employment. However, the rapid global advancement of artificial intelligence (AI) presents both transformative opportunities and substantial challenges for these traditionally managed enterprises, particularly in contexts marked by limited digital infrastructure and strong cultural emphasis on intergenerational control.

This thesis investigates the contextual barriers, enablers, and optimal strategies for AI adoption in Cambodian family businesses. Employing a mixed-methods explanatory sequential design, the study combines a survey of middle to senior management with in-depth semi-structured interviews with 10 business leaders/owners

, grounded in the Technology Acceptance Model (TAM) and the Technology-Organization-Environment (TOE) framework.

Expected findings will identify key obstacles such as low digital literacy and resource constraints, alongside enablers like competitive pressures and government digital initiatives. The research anticipates proposing phased, context-sensitive AI integration strategies focused on high-impact areas like customer relationship management and supply chain optimization.

These outcomes will advance theoretical understanding of technology adoption in emerging-market family firms and offer practical, actionable guidance for business leaders and policymakers to foster inclusive digital transformation, enhancing competitiveness and long-term sustainability of Cambodia's vital family business sector.

Keywords

AI adoption; Cambodian family businesses; barriers to AI; enablers of AI; contextualized strategies; emerging markets; family-owned enterprises; Technology Acceptance Model (TAM); Technology-Organization-Environment (TOE) framework; mixed-methods research; digital transformation; SME AI integration

Chapter 1 Introduction

The In an era where artificial intelligence (AI) is reshaping industries worldwide, family businesses face a pivotal crossroads. These enterprises, often deeply rooted in tradition, generational leadership, and cultural values, must navigate the integration of cutting-edge technologies to remain competitive. Yet, in developing regions like Southeast Asia, AI adoption remains uneven, hampered by unique challenges such as limited resources, resistance to change, and infrastructural gaps. This disparity is particularly pronounced in Cambodia, where family firms dominate the economic landscape, contributing significantly to sectors like manufacturing, agriculture, and service, but lag behind in digital transformation. Addressing these issues is crucial because AI holds the potential to drive efficiency, innovation, and sustainable growth, enabling family businesses to preserve their legacies while adapting to global markets. Without targeted strategies, however, these firms risk obsolescence, exacerbating economic inequalities and stalling national development.

This thesis explores the barriers and enablers to AI adoption specifically within Cambodian family businesses, with a focus on developing contextualized strategies tailored to their socio-cultural and economic realities. The central research question guiding this work is: What are the key barriers and enablers influencing AI adoption in Cambodian family businesses, and how can contextualized strategies overcome these to foster successful integration? This inquiry is delimited to large family businesses in Cambodia, drawing on qualitative and quantitative insights from existing literature and empirical data, while excluding non-family-owned entities. By concentrating on this niche, the study aims to provide practical, region-specific recommendations rather than broad generalizations.

The outcomes of this research are expected to represent a significant advance in the knowledge domain of family business management and digital innovation in emerging markets. While global and Asian studies have highlighted general barriers like socioemotional wealth concerns and generational resistance, little attention has been paid to Cambodia's unique context, including its nascent digital infrastructure and post-conflict economic recovery. This thesis bridges that gap by offering evidence-based strategies that could enhance AI uptake, potentially serving as a model for similar economies in Southeast Asia and contributing to broader discussions on inclusive technological advancement.

The remainder of this thesis is structured as follows. Chapter 2 reviews the existing literature on AI adoption in family businesses, synthesizing global, Asian, and Cambodian perspectives to identify key themes in barriers and enablers. Chapter 3 outlines the methodology, including data collection approaches such as surveys, and interviews, focused on Cambodian family firms. Chapter 4 presents the work plan and Implication.

Chapter 2 Literature Review

The Artificial intelligence (AI) refers to computational systems and advanced algorithms capable of performing complex tasks that traditionally require human intelligence, including reasoning, learning from data, predicting outcomes, and adapting to new environments (Berente et al., 2021). At its core, AI encompasses technologies like machine learning, natural language processing, and robotics, enabling machines to process vast amounts of information and make decisions autonomously (von Krogh, 2018). This capability positions AI as a critical factor for firms seeking to develop a sustainable competitive advantage in the digital economy (Gregory et al., 2021; Kemp, 2024; Kumar et al., 2025). For businesses, AI offers numerous advantages, such as enhancing operational efficiency through automation of routine tasks, improving decision-making with data-driven insights, and fostering innovation by uncovering patterns in large datasets that humans might overlook (Gregory et al., 2021). In family businesses, these benefits can translate to better resource allocation, personalized customer experiences, and long-term growth, allowing firms to preserve their legacies while scaling operations in competitive markets (Soluk et al., 2025). For instance, AI can optimize supply chains, predict market trends, and boost productivity, ultimately leading to cost savings and revenue growth (Hendriksen, 2023).

However, alongside these advantages, AI adoption comes with significant disadvantages that can complicate its integration, particularly in traditional organizational structures (Raisch & Krakowski, 2021). The core traits of AI, such as autonomy, learning, and inscrutability, may “alter organizations in unprecedented ways,” posing important challenges, particularly for family firms, due to families' desire to retain control and a unique culture that is mostly inward focused and described as change reluctant (Berente et al., 2021; von Krogh, 2018; Diaz-Moriana et al., 2020; König et al., 2013; Kotlar et al., 2013; Glikson & Woolley, 2020). AI-related challenges also include managers' fears about losing their jobs and owners' fears about losing control, the risk of disintermediation, and issues concerning data privacy and security (Raisch & Krakowski, 2021; Vanneste & Puranam, 2024; Chalmers et al., 2021; Laine et al., 2024). Moreover, there are inherent conflicts of AI adoption in firms' supply chains, for instance, the potential to disrupt established and proven processes and relationships (Hendriksen, 2023; Mithas et al., 2022). These and related challenges make the adoption of AI in family firms a particularly complex and knowledge-intensive endeavor (von Krogh, 2018). Research indicates that the complexity and knowledge intensity of AI underscore the need for careful implementation, as mismatches with existing systems or inadequate skills can lead to failed investments and heightened vulnerabilities (Berente et al., 2021; Raisch & Krakowski, 2021).

This tension between AI's transformative potential and its inherent risks is especially evident in family businesses, which globally exhibit a cautious approach to adoption, with rates often below 5% in key regions (Worek & Aaltonen, 2025; Soluk et al., 2025). Rooted in socioemotional wealth (SEW) theory, family firms prioritize non-financial goals like reputation and harmony, fostering conservatism

toward disruptive technologies (Gómez-Mejía et al., 2007; Berrone et al., 2012; Kellermanns et al., 2012). Soluk et al. (2025) highlight how external networks with suppliers, customers, and competitors can bridge knowledge gaps, yet passive family ownership, holding equity without hands-on management, amplifies obstacles like fears of autonomy loss, while active family management leverages these ties to build trust and facilitate integration (Soluk et al., 2025). Ulrich et al. (2023) further note that family influence correlates with lower AI openness, as owners view it as a threat to emotional stakes, though SMEs may benefit from agile decisions (Ulrich et al., 2023). Worek and Aaltonen (2025) frame these issues through the Technology-Organization-Environment model, emphasizing regulatory hurdles but suggesting alignment with family values can enable growth (Worek & Aaltonen, 2025). Buçaj (2024) advocates for strategies addressing generational resistance and skill gaps, collectively illustrating AI's capacity to enhance efficiency while pinpointing barriers like workforce anxieties, privacy concerns, and cultural shifts (Buçaj, 2024; Soluk et al., 2025; Ulrich et al., 2023).

In Asian contexts, where family businesses drive 60-70% of GDP, cultural legacies intensify this clash, leading to technological inertia (Teixeira et al., 2020; Khan, 1999). Teixeira et al. (2020) offer a systematic review of Asian family business research, identifying succession, governance, and innovation as central motifs, with AI uptake quickened by generational changes but bogged down by subpar data and training expenses (Teixeira et al., 2020). Li et al. (2025) cleverly use China's One-Child Policy reform as a natural experiment to illustrate how more heirs cultivate forward-looking strategies, spurring AI investments to secure succession and thrive in tough markets (Li et al., 2025). This idea aligns with Wang et al. (2025), who examine data from Chinese A-share listed family firms spanning 2007 to 2024, concluding that AI fuels innovation by heightening risk tolerance and streamlining resources, with the strongest impacts in smaller, family-founded companies, those with deep management involvement, and firms on the cusp of generational handovers (Wang et al., 2025).

Turning to Cambodia, family businesses form the backbone of the economy, particularly through small and medium-sized enterprises (SMEs) that dominate sectors like manufacturing, agriculture, and services, contributing significantly to job creation, household income stability, and overall growth (Sukamdani, 2023; The experiences and extent of adopting AI among SMEs in Southeast Asia, 2025). As major employers and innovators, these firms stimulate local economies by fostering entrepreneurship, supporting community development, and adapting to post-conflict recovery needs, with studies highlighting their role in financial sustainability and marketing performance (e.g., through successor skills in family-owned enterprises) (Teixeira et al., 2020; Young, n.d.). Family businesses in Cambodia often co-managed by couples or generations provide stable employment, enhance household financial literacy, and drive inclusive growth, making them vital for poverty reduction and economic diversification (Sukamdani, 2023; Stewardship Asia, n.d.).

Despite these strengths, there are no direct studies on AI adoption in Cambodian family businesses, revealing a critical research gap (Teixeira et al., 2020; The

experiences and extent of adopting AI among SMEs in Southeast Asia, 2025). This thesis innovates by exploring these links in Cambodia's unique post-conflict landscape, developing contextualized strategies to harness AI for sustained family firm success and broader economic advancement (Soluk et al., 2025; Wang et al., 2025).

Chapter 3 Research Methods

The This study employs a mixed-methods research design with an explanatory sequential approach to examine barriers, enablers, and contextualized strategies for AI adoption in Cambodian family businesses. The mixed-methods choice addresses the phenomenon's complexity, where quantitative data offers breadth and generalizability on prevalence across samples, while qualitative data provides depth and contextual insights into unique dynamics. This builds on literature emphasizing the need for nuanced exploration in emerging markets, developing from global surveys but diverging to incorporate Cambodia-specific cultural and infrastructural factors.

The research follows a two-phase design. In the quantitative phase, a structured questionnaire, developed using the Technology-Organization-Environment (TOE) framework and Technology Acceptance Model (TAM), is administered to 10 large family businesses, proportionally allocated based on Cambodia's enterprise structure. The survey targets middle- and high-level executives, as they handle operational aspects and can report on practical challenges like resource constraints and perceptions of usefulness/ease of use. Responses are collected via a five-point Likert scale to quantify barriers and enablers, etc. In the qualitative phase, 10 semi-structured interviews are conducted with top management leaders/owners to build on and explain survey findings, exploring themes like governance, succession, and cultural readiness through open-ended questions.

The study is situated in Cambodia, focusing on agriculture, manufacturing, and services sectors. Data analysis uses Stata for the quantitative phase, including cleaning, descriptive statistics (means, frequencies), and inferential techniques (correlation, regression) to identify relationships. Qualitative data undergoes thematic analysis, with inductive and deductive coding to capture emergent insights aligned with TOE/TAM.

Chapter 4 Work Plan and Implications

Given the pronounced scarcity of empirical data on family businesses in Cambodia, particularly with respect to AI adoption, the process of primary data collection, rigorous analysis, and publication in peer-reviewed journals will inevitably require additional time to meet the exacting standards of academic scrutiny. Nevertheless, this doctoral research carries substantial significance by illuminating an underexplored context within post-conflict emerging markets,

thereby addressing a critical knowledge gap in the literature on technology adoption among family firms. The anticipated findings are positioned to inform evidence-based policymaking that fosters inclusive digital development, bolsters enterprise competitiveness and intergenerational sustainability, and supports broader objectives of poverty alleviation and economic diversification, offering valuable guidance to government agencies, policymakers, business associations, and international development organizations committed to advancing digital inclusion in Cambodia and analogous emerging markets. Accordingly, every effort will be made to complete and disseminate the required publications as expeditiously as feasible, ensuring that the thesis and its associated outputs achieve high scholarly standards, relevance, and impact. Should the proposal be successfully defended in January 2026, the subsequent five to six months will be devoted to finalizing and submitting the remaining Scopus-indexed publication, while concurrently synthesizing the full body of results into the completed thesis manuscript, with the aim of being prepared for the final defense in November or December 2026, subject to supervisory approval. The overarching purpose of this structured yet flexible plan is to maximize the research's scholarly and practical impact while sharing timely, context-sensitive insights with academic, policy, and practitioner communities.